Power supply

File: power_supply.kicad_sch

Camera LEDs

File: camera_leds.kicad_sch

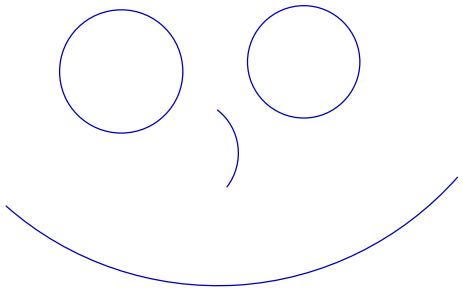
Amplifier

File: Amplifier_Circuit.kicad_sch

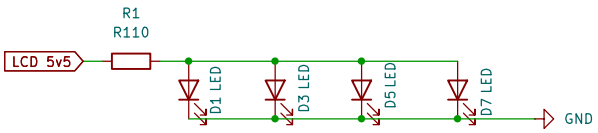
Hand Controller

File: hand_controller.kicad_sch

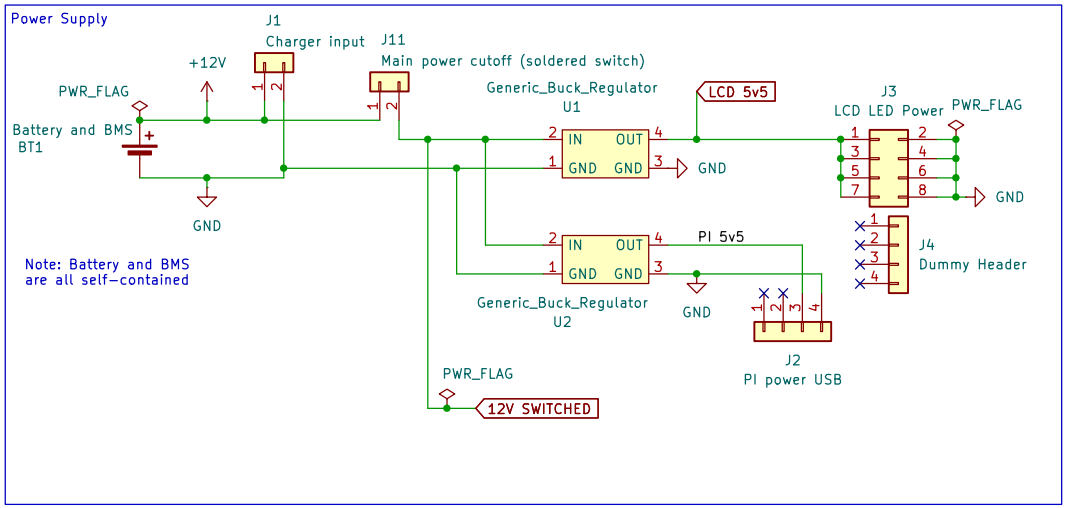
H1
MountingHole
H2
MountingHole



Camera illumination

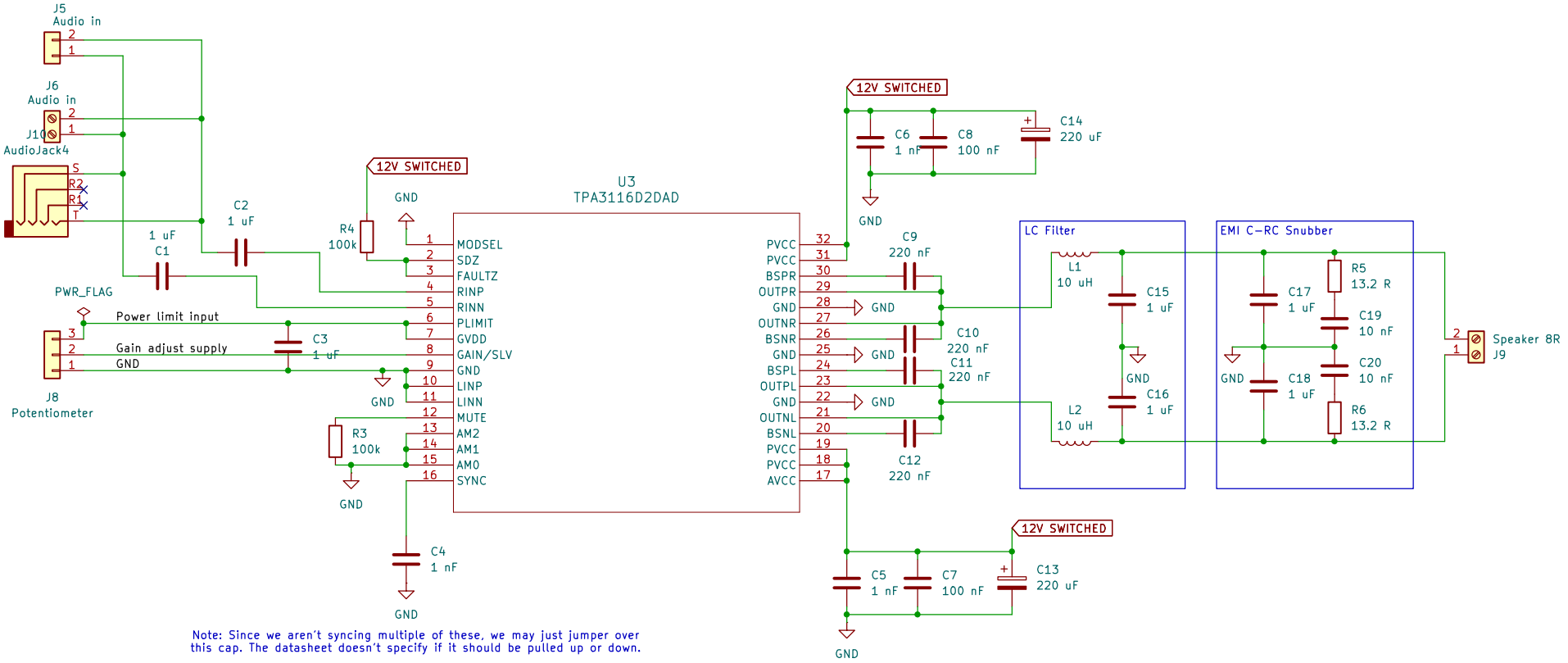


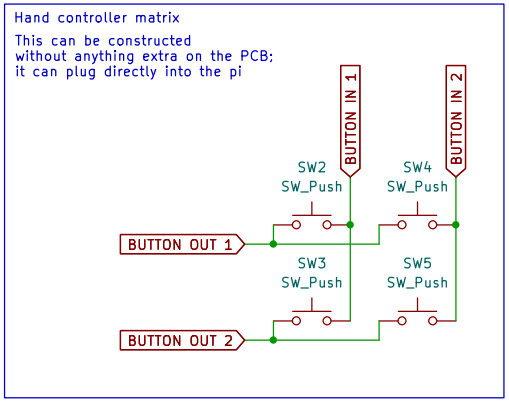
We're leeching off the LCD power supplies;
they have enough margin for this


Power Supply
J1
Charger input
J11
Main power cutoff (soldered switch)
+12V
PWR_FLAG
Battery and BMS
BT1
GND
Note: Battery and BMS
are all self-contained
Generic_Buck_Regulator
U1
IN
OUT
GND
GND
LCD 5v5
J3
LCD LED Power
PWR_FLAG
GND
J4
Dummy Header
PI 5v5
Generic_Buck_Regulator
U2
GND
J2
PI power USB
PWR_FLAG
12V SWITCHED

TA recommends 0805 SMD, 0402 is a bit smaller, 0603 is the smallest he'd recommend

Multiple audio input options based on what we need:
- AUX jack
- screw terminal
- hard-wired




Hand controller matrix
This can be constructed
without anything extra on the PCB;
it can plug directly into the pi
BUTTON IN 1
BUTTON IN 2
SW2
SW_Push
SW4
SW_Push
BUTTON OUT 1
SW3
SW_Push
SW5
SW_Push
BUTTON OUT 2